

Preregistration document for the color-discrimination project with different adapting points

Fangfang Hong¹, Cameron Zhang¹, Kara Wang¹, Jason Chow², Alex H. Williams^{3,4}, Phillip Guan², David H. Brainard¹

¹University of Pennsylvania, Department of Psychology

²Reality Lab Research, Meta

³New York University, Center of Neural Science

⁴Flatiron Center for Computational Neuroscience

More Trials

After examining the comparison between extended WPPM fit to the sparsely sampled conditions and the WPPM fit to each of the densely sampled conditions with the data as preregistered in 2026-04-08 addendum, we concluded that we may continue to under-try in the sparsely sampled conditions, in that the agreement was not as good as in the simulations. A possible reason for this is that trial placement in the real experiment is not as efficient as in the simulations, because of the need to run fallback trials in the real experiment. Because our participants remain available as we consider the question in more detail, we have decided to run an additional 500 trials per sparsely sampled condition, continuing on from where we left off as described in the previous (2026-04-08) preregistration. This will require approximately 10 more sessions per participant. We will then make the same comparison as described in the previous document with the additional trials, and also consider other ways (e.g. after contrast normalization) of analyzing the data. If we still consider that it is possible we are under-tryed, we will consider using the fits of the data collected to date to refine the placement of the fallback trials in additional data collection.